

Partitioning New Disk Drives

When adding a new storage device to Linux, there are some operations that must be performed before putting the storage device into use. There is a "one time" function of creating the partition table that may not have been done on the new storage. The other "one time" function would be if you wished to change the partition scheme, to perhaps using GPT. Changing and creating the partition table can be done with the GUI using the **disks** program included in GNOME, or one of the command line programs like **sfdisk** or **fdisk**. First, we must determine if there is a partition table on the disk.

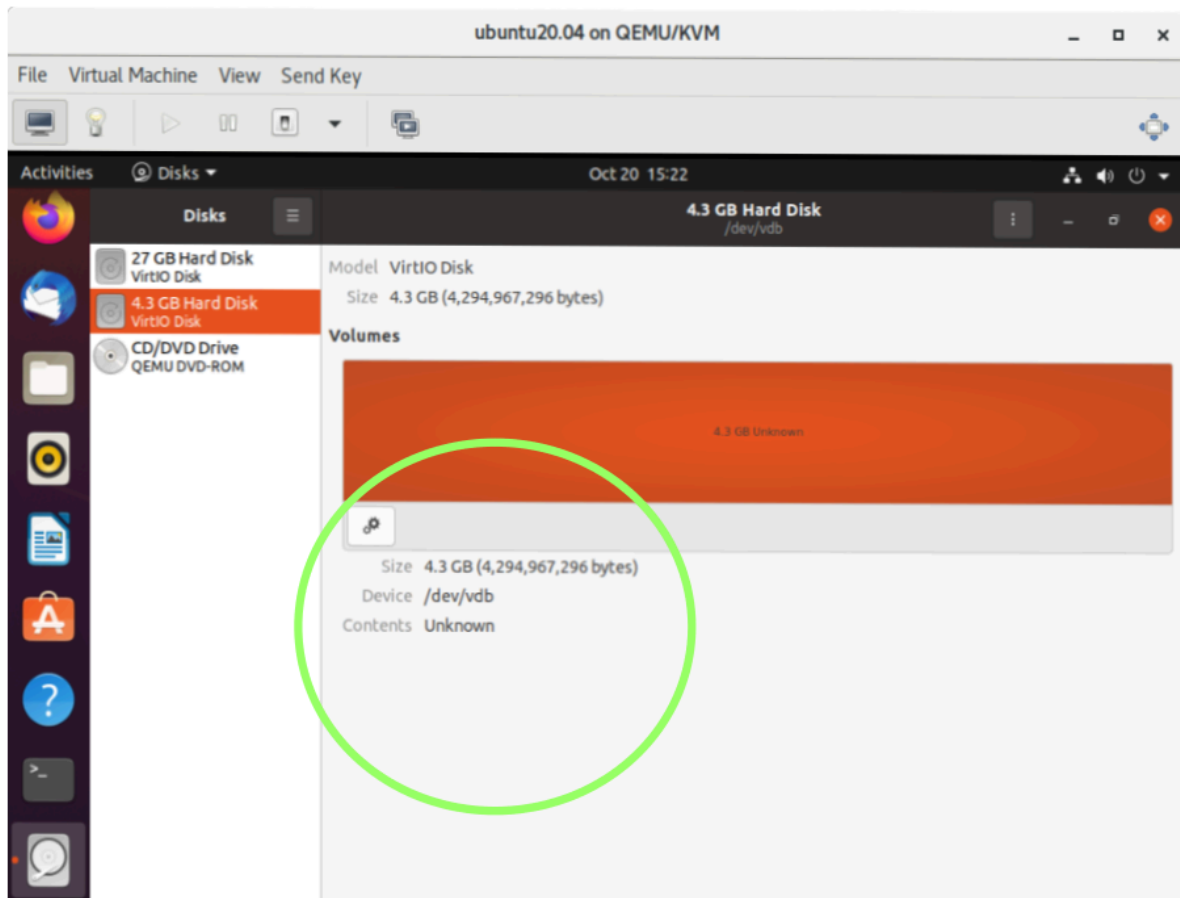
NOTE: Creating a new partition table on a disk that already has one will destroy the contents of the disk.

Our recently installed system has had a new 4GB disk drive added; to verify there is a partition table, open the GNOME **disks** application by selecting the **Applications** menu, then select **disks**.

If the **disks** program was not found, you can always type "disks" into the search window.

On the left column, select the 4GB disk. The information about the disk should populate the right column.

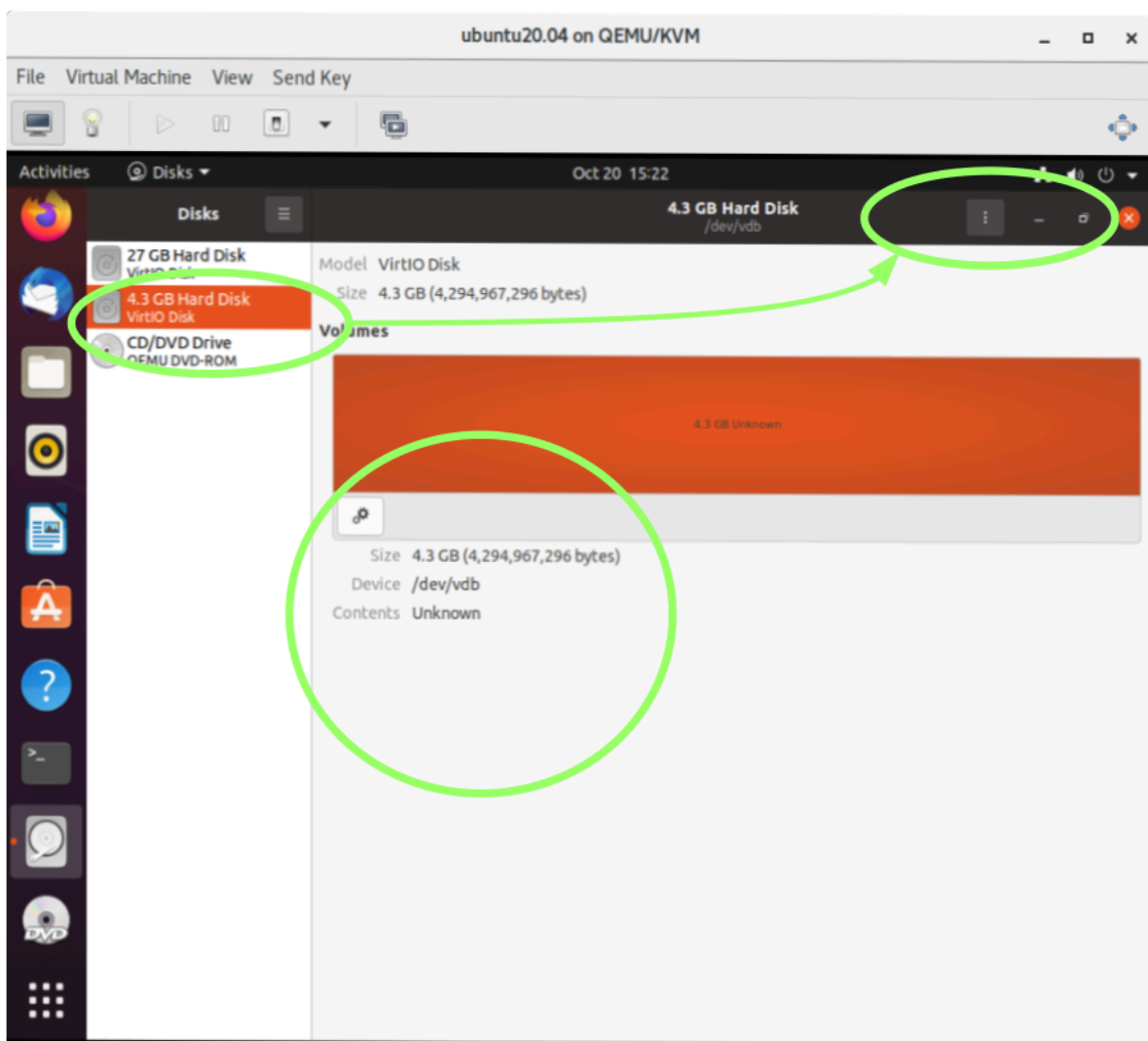
Here is an example of a disk with no usable partition table; notice the lack of detail and absence of the partition table. There are a couple of different methods to deal with the missing partition; options include the Gnome **disk** utility, **fdisk**, **gparted**, and more.





Disk With No Partition Table

Using the Gnome **disks** application, select the disk to partition, then open the menu on the top right to expose the choices.

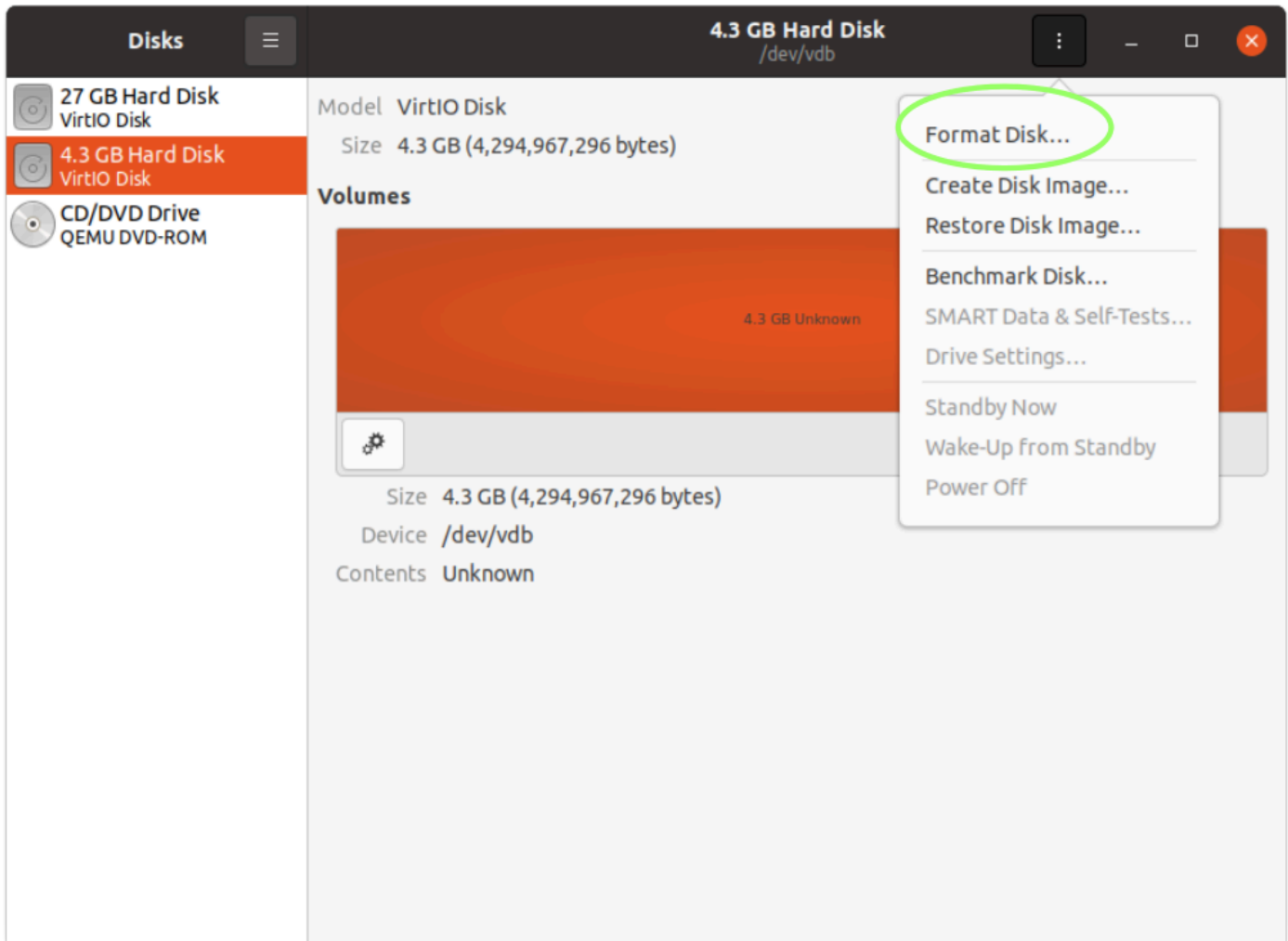


Display the Location of the Format Menu

NOTE: In the next few screenshots, the Gnome **disks** program uses the term "volume" for partitions and whole disks; proceed carefully.

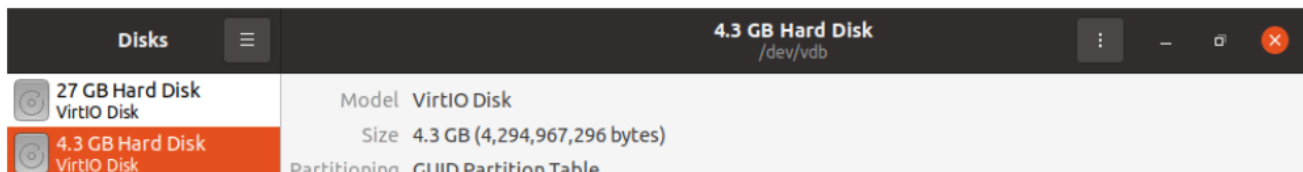
whole disks, proceed carefully.

Once the menu is expanded, choose the **Format Disk** option. Be very careful that the correct disk is selected on the left side of the screen, as the next operation can DESTROY the disk selected. Unless otherwise directed, the formatting defaults are fine.



Display the Volume Level Parameters

Once the partition table has been written, one or more partitions may be created. The + now appears on the hard disk detail screen. Use the + to add partitions. **Volumes** refers to the whole disk here.



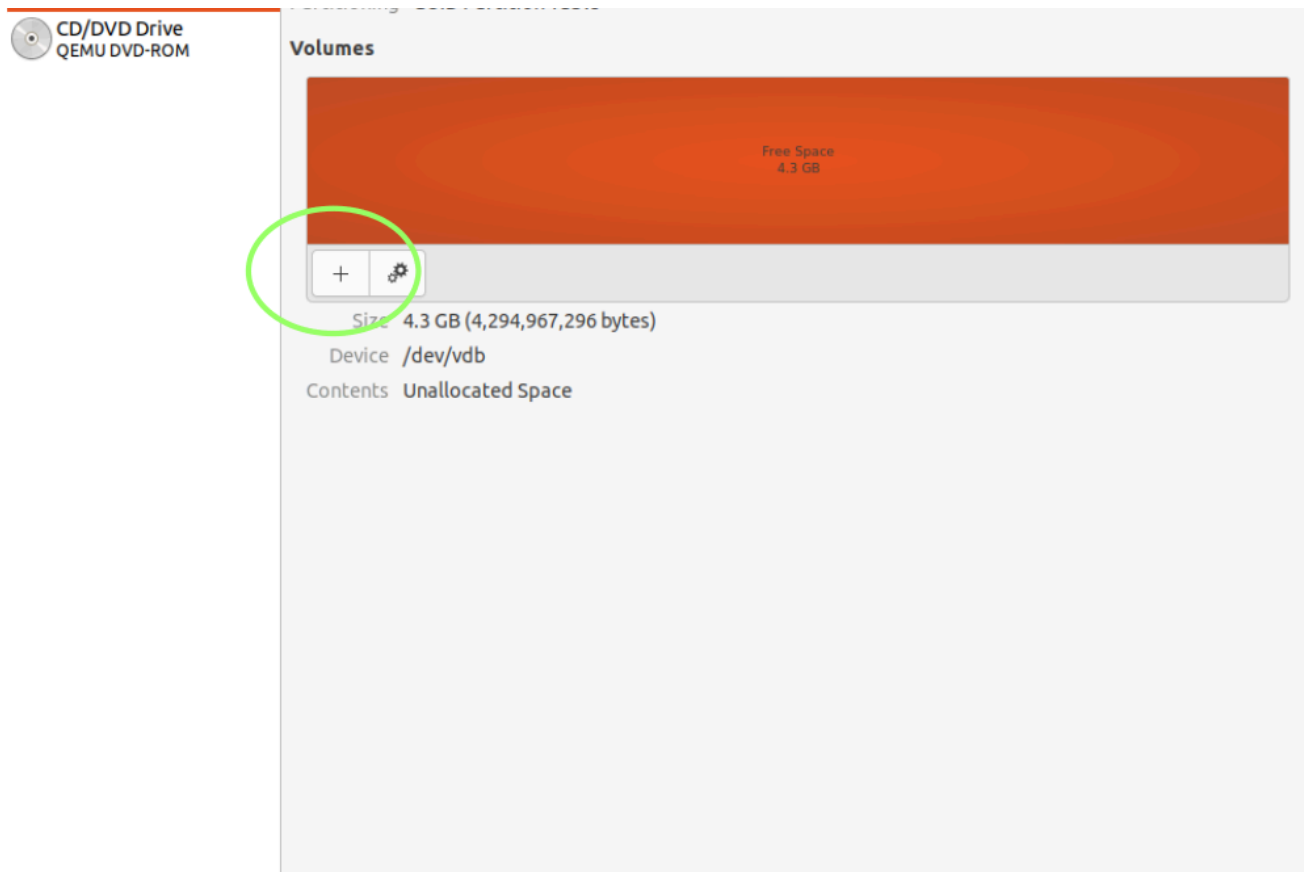
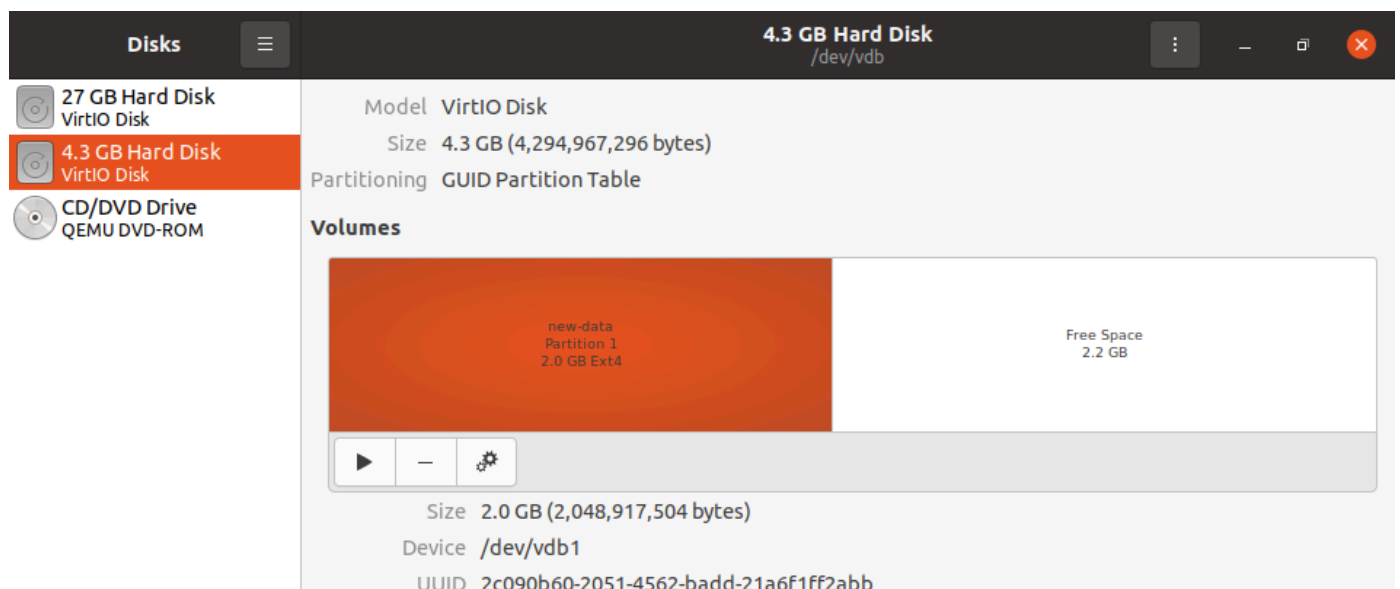
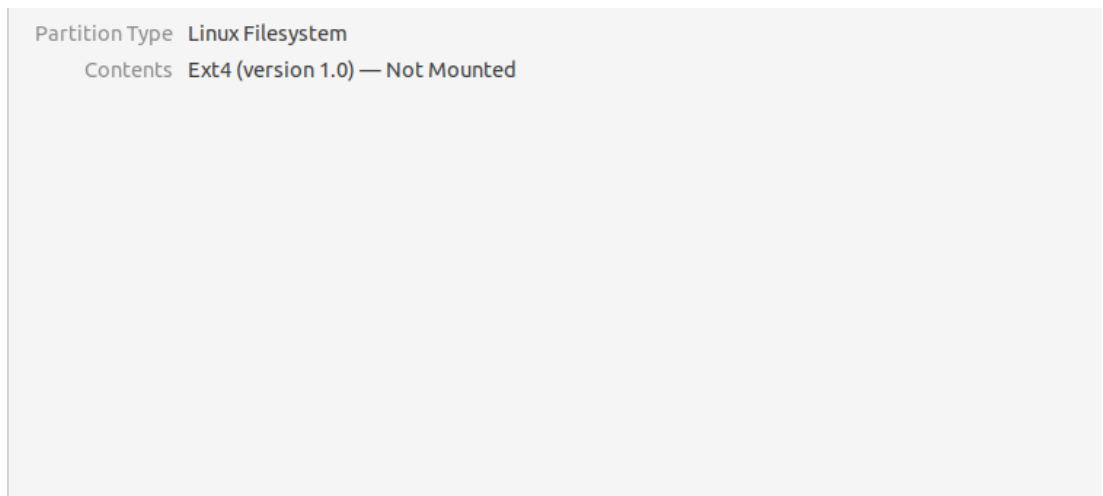


Image of Disk After Being Formatted

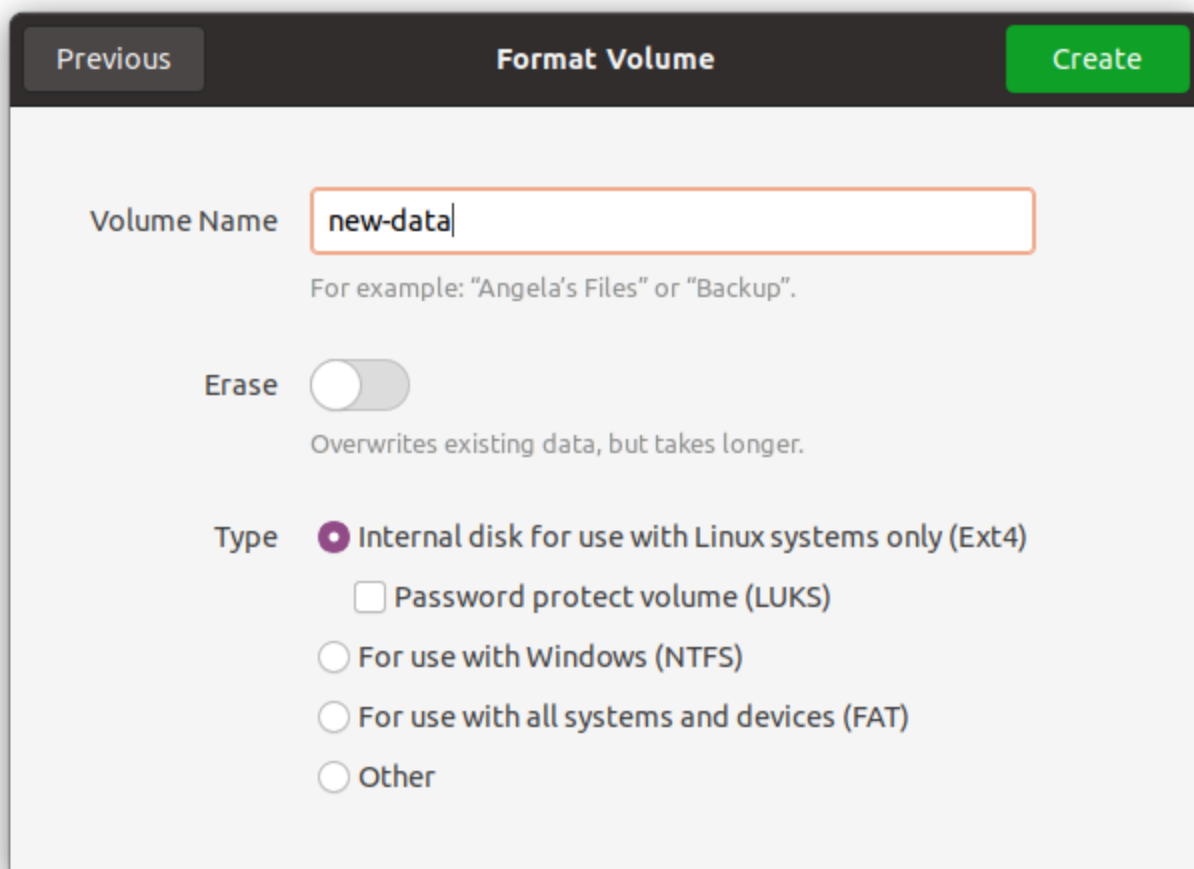
Select the size of the partition. The default is to use the entire disk. For our testing example, use half of the disk, saving space to experiment. The term **Volumes** refers to the partition(s) here.





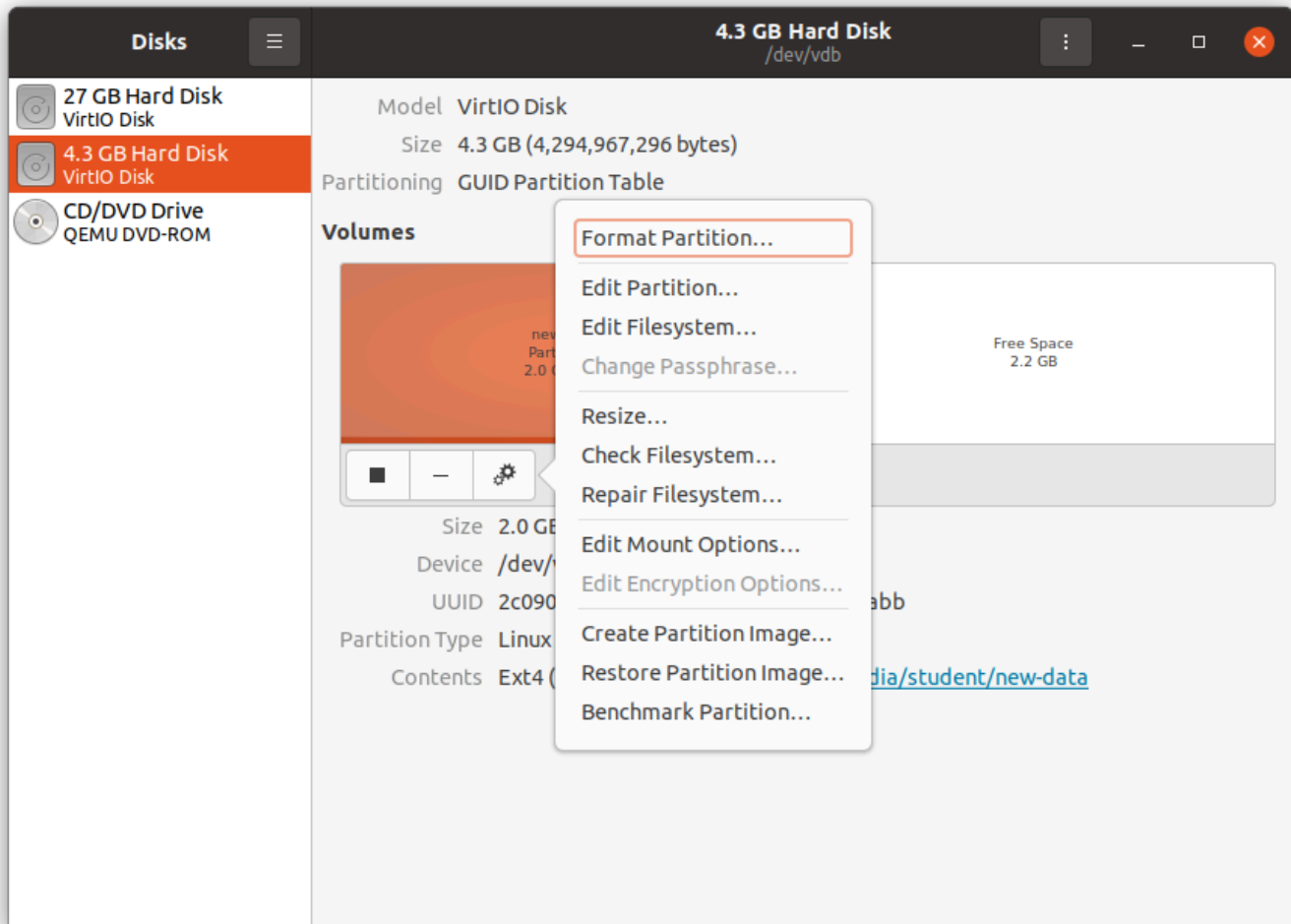
Display the Completed Addition of a Partition

Select **Next** to view or change the information used to create the partition table. We added a label (alias name) to the partition. This is also where the filesystem type is selected.



Format Volume

Once the partition is formatted, additional options are available to manipulate the partition, such as resize, change mount points and, of course, format partition. Formatting a partition prepares it to be used as a filesystem in Linux.



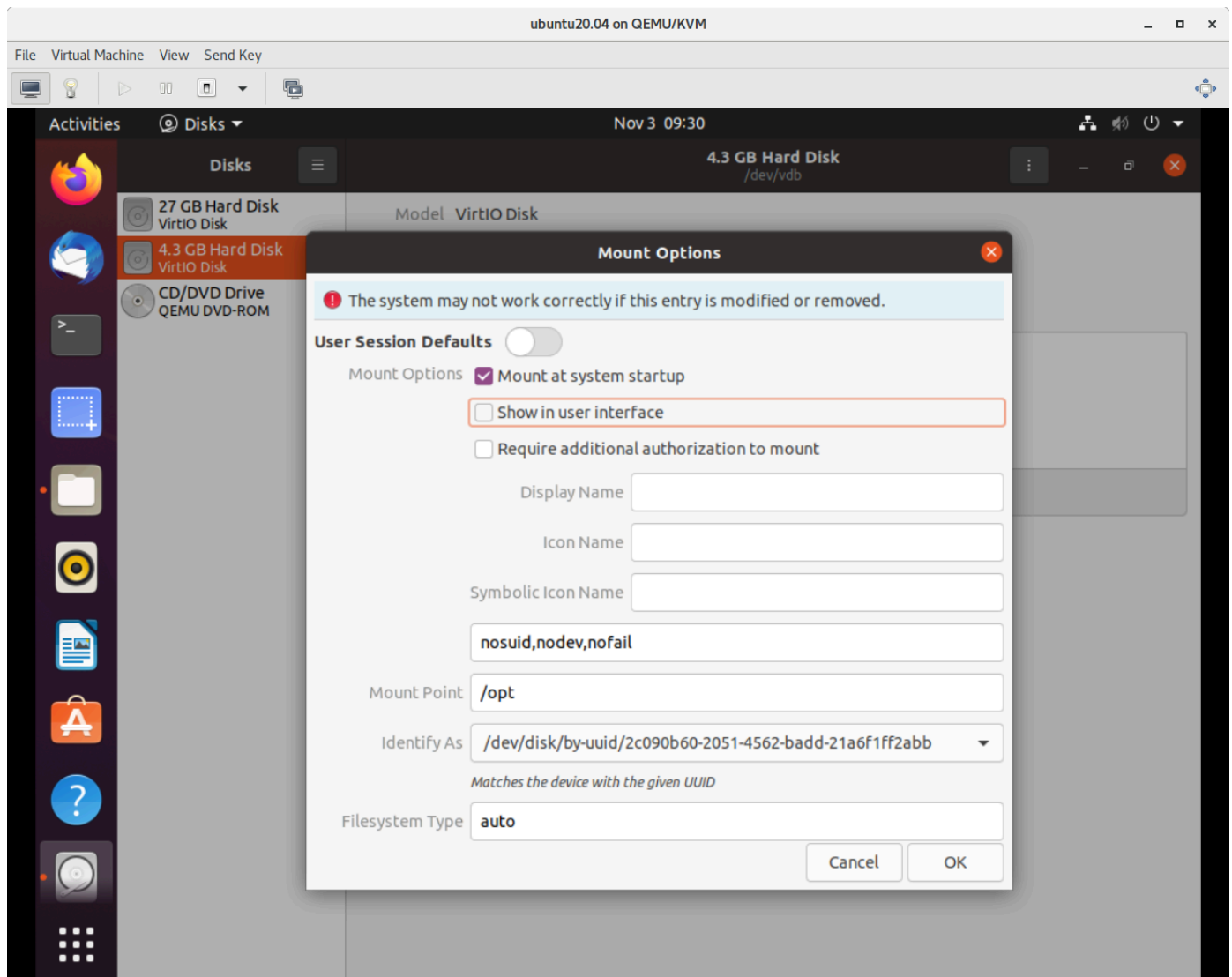
Additional Options

One of the available options is ***Edit Mount Options***, which can change where in the filesystem tree structure the filesystem is attached.

To use the newly created filesystem, or any filesystem, it has to be mounted first. The action of mounting a filesystem overlays a directory inode with the inode from the newly formatted partition. The program that connects one filesystem to another is called **mount**; to disconnect the partition when

finished, you use `umount` (not a typo; there is no "n"). The options in the **gnome-disks** application will facilitate naming the location of the new mount within the directory structure, mounting the new filesystem, and setting these actions to be permanent.

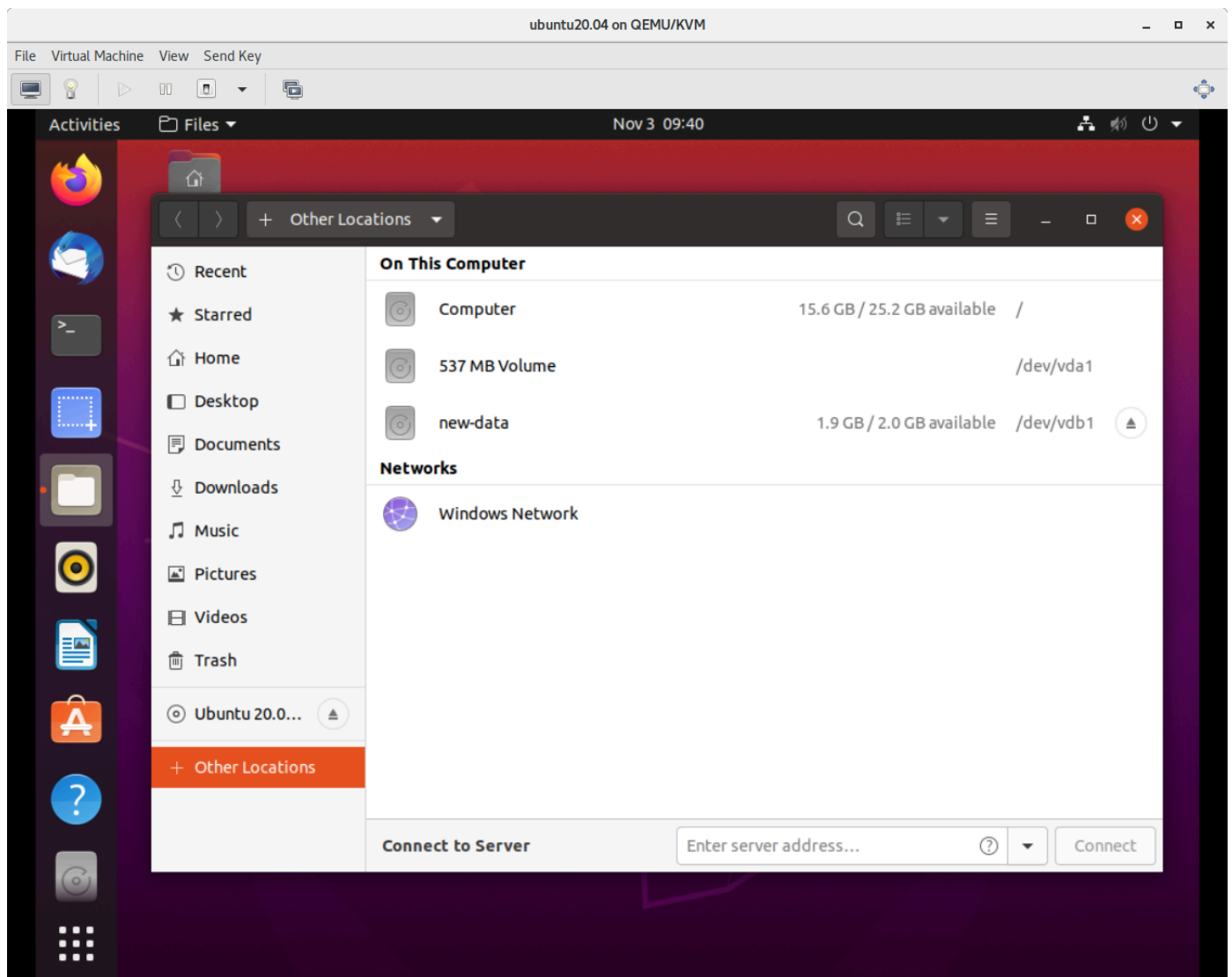
Change the mount options so that the new partition is mounted on `/opt` automatically at boot time. Click **OK**.



Mount Options

After rebooting the system, the new filesystem should be mounted and available. There are several ways to test if a filesystem is mounted; the command `df` shows the mounted partitions.

The file manager will also show mounted filesystems under the **Other Locations** tab.



Other Locations Tab